CRYSTALLOGRAPHY

BY

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RESEARCH AND THEORY

A PUBLISHED CELL EDGE READ BACK FROM EXACT SITE COORDINATES

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For Megan, my loving wife, the only person that really understands what I mean when I speak.

For Autumn and Opal, without you this thought would have never occurred.

To everyone else, I wish you well.

ABSTRACT

A crystal carries an answer somebody else wrote down in advance. No other control in this work does.

Tile a published unit cell, carry every site as an exact integer, and score how well the arrangement agrees with itself shifted along one axis. The offset where agreement peaks is the cell edge. Across 453 axes drawn from the Crystallography Open Database, every recovered period equals the published edge as an integer, with no tolerance applied and no error left to average.

The same corpus read on a grid of 0.25 angstroms, with the tiling capped at 320 voxels along the longest side, lands inside one voxel on every axis and on the published edge on none. Both bounds come from the reader and neither comes from the deposit, and they cost a mean absolute error of 0.0124 angstroms together with a bias of −0.0067. The exact reading has a signed error of zero, which places the bias entirely in the voxel floor. The cap also refuses large cells outright, a cost that never appears as error because those entries go unread.

One correction here is not about the reader. A lattice of period P agrees with itself as well at $2P$ and at $3P$, and taking the tallest lag reports a harmonic: read that way, published edges come back at almost exactly twice their value on ten of eighteen axes. Scoring a candidate together with its multiples fixes it, and the rule holds with or without a grid, because it repairs a property of periodic sets and not damage the reader does.

What this does not license is stated with it. The reading recovers the number the deposit carries and stops there, only right-angled cells are used, and exact agreement does not make the published edge true. The interest is not the cell edge, which is published to more decimal places than anything here reaches. The interest is that a memoryless control can only show an instrument does not invent structure, and a crystal lets it show that it finds structure that is there.

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1

WHAT THIS IS

This chapter states the construction, places it beside the crystallographic object it resembles, and marks the boundary of what the work can claim about it.

1.1 THE CONSTRUCTION

Tile a published unit cell, carry every site as an exact integer, and score how well the arrangement agrees with itself shifted by each candidate offset along one axis. The offset where agreement peaks is the cell edge.

That is an autocorrelation. For one lag along one axis, the score is the share of *occupied* sites whose neighbor at that distance carries the same value. The candidate lags are swept and the score is read at each.

Empty positions are left out, and the measurement depends on that exclusion. A structure built from atom sites is mostly empty space, and counting the empty positions would report how empty the structure is, which every structure agrees on.

1.2 THE OBJECT IT RESEMBLES

The autocorrelation of electron density in a crystal is the Patterson function [2], whose peaks sit at the vectors between atoms. The construction here is a one-axis, element-labeled relative of it: it is read on sites and not on a density, two sites agree only when they carry the same element, and the cell edge is the lattice vector at which every site agrees.

No priority is claimed for the construction. It is reached from a general question, how far an arrangement sits from the max disordered arrangement of its own parts, with no crystal in mind. Arriving from there at an object a field already uses is evidence that the general question is pointed at something real. A full placement of this construction in the crystallographic literature is not made here.

1.3 WHY THE CRYSTAL IS HERE AS A CONTROL

The interest is not the cell edge. Cell edges are published, refined to more decimal places than anything here reaches, and recovering one adds nothing to crystallography.

The interest is that the answer is known in advance by somebody else. Apart from the protein structures, every other control in this work is a memoryless process, and a memoryless control

shows only that an instrument does not invent structure. It cannot show that the instrument finds structure that is there. Those are different claims, and an instrument has to earn the second.

A crystal earns it. The periodicity is not inferred from the measurement; it is a fact about the mineral, and the Crystallography Open Database [1] publishes the cell for every entry it holds. The number the detector has to hit is written down by people with no interest in whether it works.

2

WHAT IS ACTUALLY MEASURED

This chapter states the pipeline from a deposited structure to a recovered edge, so that a reader can say what the number is a number about.

2.1 FROM A DEPOSITED CELL TO EXACT POINTS

A CIF entry gives the cell lengths, the angles, and the fractional coordinates of the sites in one cell. Four steps turn that into something the detector reads.

1. **The angles are checked.** Only cells with right angles are used. An oblique cell would need a shear before a shift along one coordinate meant a shift along an axis, and that shear is a second thing able to go wrong between the published number and the recovered one.

2. **The cell is tiled.** A single cell holds one period and one period cannot be told from noise. Four tiles an axis, and there is no cap on this: the points are sparse. The cost is linear in the tile count and not the cube of a grid side.

3. **The sites are carried as exact integers.** A deposit writes decimal text, and that text is a numerator and a count of decimal places, both integers. A fractional coordinate plus a whole tile, times a cell edge, is a product of two decimals carried through as integers. Nothing rounds. There is neither grid, voxel nor nearest anything.

4. **Each axis is asked separately** for its period, and all three answers are retained. One structure therefore contributes up to three independent results.

Nothing at any step tells the detector what it is looking at. It receives positions carrying values and an axis index.

A grid is a cost the deposit does not impose. Assigning sites to voxels of 0.25 angstroms and capping the tiling at 320 voxels along the longest side are both bounds from the reader and neither comes from the deposit. The evidence chapter measures what they cost: a mean absolute error of 0.0124 angstroms on a measurement whose error is otherwise zero. The cap also refuses large cells outright, a second cost that never appears as error because those entries go unread.

2.2 READING AN AXIS WITHOUT AN ORDER

An axis is read as a lookup from a coordinate to the arrangement located at it. Two coordinates agree when their arrangements match, values included, and a period is a coordinate whose arrangement matches the one exactly a period away.

The values have to be in the comparison. Leaving them out reads a rocksalt cell at half its published edge, correctly: the two sublattices interleave. The positions alone do repeat every $a/2$ and only the elements distinguish the two halves.

The candidates are every difference between two occupied coordinates, and that set is complete. A period that agrees with anything at all is by construction a difference between two things that agree. It is in the set. There is neither sweep, ceiling nor stride, and nothing in the measure narrows what can be found.

Membership is the only relation asked of the set. The search is a lookup. A binary search over sorted coordinates answers the same question and imposes an order the set does not have, paying a full comparison at every step of it.

Taking the tallest lag is wrong. A lattice of period P agrees with itself just as well at $2P$ and at $3P$. Which of those is tallest is settled by noise. The tallest lag reports a harmonic and not the fundamental. Read that way, published cell edges come back at almost exactly twice their value on ten of eighteen axes.

The correction is to score a candidate on itself together with its multiples, and to require at least two members of that family to be present before the candidate is scored. Exactly two members are used and never more: a family growing as the candidate shrinks lets a short wrong candidate win by holding more members, since it only has to catch one good lag among them. An uncapped family reads herzenbergite, molybdite and one more mineral short by a factor near two thirds.

The family rule holds in the exact reading as it does on a grid. The grid's other costs are damage done by the reader. This one is a property of periodic sets themselves: a set with period P agrees with itself at $2P$ whether anybody rounds it.

This is the harmonic trap, and it is not specific to crystals. Any reading of a period by its tallest peak falls into it. Scoring against multiples instead of taking the tallest peak is the general fix, and the crystal case is where it is easiest to see, because the right answer is published and a factor of two is unmistakable.

2.3 WHAT A FAILURE WOULD LOOK LIKE

The criterion is stated before the numbers, since afterwards it is worth nothing.

A recovered period not equal to the published edge would mean the detector is not reading the period. A near miss counts as a miss, since nothing in this path rounds and there is no tolerance to fall inside. A systematic bias in one direction would mean it is reading the tiling and not the crystal, since the tiling is the only structure present that the mineral did not put there. Both are visible in the reported columns without knowing which entry is which.

A gridded reading has to set that bar at one voxel, because it has a voxel. It passes 453 of 453 and carries a bias of -0.0067 angstroms. The exact reading passes the stricter test and has no bias to explain.

3

THE EVIDENCE

This chapter reports every number the sweep produces, the exact reading beside the same corpus read on a grid, and what the comparison settles.

3.1 WHAT WAS SWEPT

Sixty four mineral names were put to the Crystallography Open Database [1] by text search. Every entry returned with right angles was fetched, tiled, and asked for its period along each of its three axes.

names searched	64
minerals that returned a usable entry	55
structures read	151
axes measured	453

One structure contributes up to three axes, and every axis is an independent question carrying its own published answer. A wrong result cannot hide in a small denominator, which a sweep buys and a sample does not. Three minerals returning three exact answers would still leave room for coincidence.

The same corpus is read twice. The gridded reading puts every site on a grid of 0.25 angstroms before the detector sees it. The exact reading carries each site as an exact integer and rounds nothing.

3.2 THE RESULT

axes equal to the published edge	453 of 453, 100.0 percent
error	zero, on every axis
tolerance applied	none

Equal as integers, with no tolerance. Every coordinate is carried at 10^{-1024} angstroms, a period is read off the whole difference set of the axis, and the comparison against the published edge is an equality between two integers. There is no tolerance to set and no error to average, because there is no error.

The failure stated in advance is any axis whose recovered period is not equal to its published edge. A near miss counts as a miss, since nothing in this path rounds. No axis misses.

3.3 THE GRIDDED READING, AND WHAT IT MEASURES

The same 453 axes, read on the grid:	
axes inside one voxel of 0.25 angstroms	453 of 453, 100.0 percent
axes equal to the published edge	0 of 453
mean absolute error	0.0124 angstroms
median absolute error	0.0108 angstroms
90th percentile	0.0243 angstroms
99th percentile	0.0404 angstroms
worst single axis	0.0554 angstroms

Read alone, that is a good result. Read beside the exact one, it says something else: the gridded reading never once lands on the published edge, and every figure in the table is the grid.

None of that error is in the deposit and none of it is in the detector. A cell edge is published to five decimal places and the detector compares integers. The 0.0124 is introduced in the reader, before the measurement begins. Reporting it as the accuracy of the instrument mistakes the reader for the instrument.

3.4 THE BIAS IS THE VOXEL FLOOR

The gridded reading carries a bias:	
mean signed error	−0.0067 angstroms
axes read long	120
axes read short	327

The detector reads short about three times as often as it reads long, and the mean sits below zero by roughly half the median absolute error. A symmetric error would put those counts near even.

The voxel floor predicts a bias of about half a voxel spread over the tiling. **The test is to remove the rounding.** The exact reading of the same 453 axes has a signed error of zero, 0 axes long and 0 axes short. The bias is the voxel floor in its entirety. Nothing of it survives when sites are not assigned to cells.

That is the strongest form the test can take. Changing the rounding rule would move the bias and leave the question of how much is rule and how much is something else. Removing the quantization removes the bias completely, which settles it.

3.5 THE FOUR WORST GRIDDED AXES

mineral	entry	axis	published	on the grid	grid error	exact
corundum	1544906	c	2.8446	2.9000	+0.0554	2.8446
herzenbergite	9008295	a	4.1480	4.1000	−0.0480	4.1480
quartz	4339381	c	15.4780	15.5250	+0.0470	15.4780
herzenbergite	9008785	b	11.1800	11.1389	−0.0411	11.1800

Herzenbergite appears twice, and it is the mineral an uncapped harmonic family reads short. It is the hardest case in the set on the grid and it is exact on both entries, as is every other axis. The exact reading has no worst axis.

The two largest grid errors have opposite signs: the worst case on the grid is not the bias but the resolution of the straddling ratio, and those are separate effects. Both are properties of a quantization the exact reading does not perform.

3.6 WHAT THIS TABLE IS NOT

It is not a measurement of any of these minerals. Every published edge in the table was measured by somebody else, to a precision this does not reach, and is the input the recovered value is graded against. The exact reading recovers the deposited number and makes no claim beyond it: agreeing with a deposit to the digit is not the same as the deposit being right, and the bracketed uncertainty a deposit carries is a quantum fixed upstream by the people who measured it.

It is also not 453 independent structures. It is 453 axes over 151 structures over 55 minerals, and several minerals contribute many entries. Treating the 453 as independent samples of anything would overstate what a sweep of 55 minerals shows.

3.7 AVAILABILITY

Each structure is a Crystallography Open Database entry, named by its entry number where the text quotes it. The detector, the exact site representation and the sweep are in the orior repository, `https://github.com/dstroy0/orior`.

4

WHERE THE PRECISION COMES FROM

This chapter explains how a gridded reading lands finer than the grid it samples, and why the exact reading has no such question to answer.

4.1 THE OBJECTION

On a grid of 0.25 angstroms, errors below that spacing are reported. A reader should stop there and ask how a measurement resolves anything finer than the spacing it samples at, because the majority of the time the answer is that it does not and the extra digits are decoration.

The objection is correct to raise and it has an answer. The mechanism is real and applies to any measurement taken on a grid. The better resolution is that the grid is not necessary.

4.2 QUANTIZATION MOVES THE PERIOD, IT DOES NOT DESTROY IT

A cell edge is almost never a whole number of voxels. On a 4.148 angstrom axis at 0.25 the period is 16.59 voxels. Successive tiles therefore do not land on the same lag: they land alternately on 16 and on 17, and the share of the agreement sitting at the upper lag carries the fraction between them.

The quantity is not lost at quantization; it is moved into the ratio between two adjacent lags. The recovered edge is the lag plus that fraction, times the voxel. Nothing is interpolated between samples that were never taken. Both lags are measured, and their relative weight is a measured quantity too.

Tiling matters beyond having more than one period. The fraction is only readable because many tiles are present and their offsets accumulate. One period gives one lag and no ratio.

Measured, that recovery works. Three cells at 10.1000, 8.6633 and 14.0574 angstroms come back at 10.1018, 8.6545 and 14.0568, and across the whole sweep 453 axes land inside a voxel at a median absolute error of 0.0108.

4.3 THE GRID IS THE READER'S

The voxel comes from the reader and not from the deposit: a CIF publishes a cell edge to five decimal places, and a grid is imposed before the detector sees it.

Carry the sites as exact integers instead and the objection dissolves. It is not answered. There is no sample spacing to beat, no straddling lags, no fraction to resolve, and no error distribution to report. The same 453 axes come back equal to their published edges.

The lesson generalizes past crystals. An instrument that answers finer than its own resolution deserves the scrutiny this chapter opens with. Sometimes the mechanism is sound, as it is here. It is still worth asking first whether the resolution was anybody's to impose.

4.4 WHAT THIS DOES NOT LICENSE

It does not license a claim that a cell edge is known to a thousand digits. The exact reading recovers the number the deposit carries and stops there. A deposited edge written as 4.76050(5) has an uncertainty of five in its last place, fixed upstream by the people who measured the crystal, and dropping that bracket is all this work does with it.

The exactness means that no second uncertainty is added underneath the first. The instrument contributes nothing to the error. Whatever error remains belongs to the deposit and can be argued about on its own terms.

5

WHERE IT STOPS

This chapter states the boundary of the claim. Nothing here is quoted past what it supports.

5.1 RIGHT ANGLES ONLY

Every entry read has a cell with right angles. Oblique cells are skipped, not measured and reported as failures. A shift along a coordinate is a shift along a crystal axis only when the axes are orthogonal, and on an oblique cell the coordinates would have to be sheared first.

That is a real restriction on coverage. It restricts which crystals are read and claims nothing about which crystals could be. Whether the same detector reads a monoclinic or triclinic cell after a shear is untested here.

5.2 TWO ATOMS ARE TWO ATOMS

On a grid of 0.25 angstroms two atoms can land on the same voxel, and the later one overwrites the earlier with nothing recovering the lost site. That is harmless for the lattice period, since a period is a property of the repeating arrangement and not of any particular site, but it means the grid is not the structure.

Exact coordinates collide only where a deposit puts two sites at the same place. Nothing is lost that the deposit does not already carry.

5.3 EXACT DOES NOT MEAN TRUE

The exact reading recovers the number a deposit carries and stops there. A cell edge written as 4.76050(5) has an uncertainty of five in its last place, fixed by the people who measured the crystal, and dropping that bracket is all this work does with it. Agreeing with a deposit to the digit is agreement with a deposit.

What the exactness rules out is a second uncertainty underneath the first. The instrument contributes nothing to the error. Whatever error remains is the deposit's and can be argued about on its own terms.

5.4 THE DETECTOR IS NOT THE MEASURE

The work carries two instruments, and reporting one as the other is a located source of confusion. The crystal result belongs to the shift agreement detector, the same one that returns an image width from a byte sequence and a key length from a Vigenère cipher. It is the instrument with external ground truth.

The null permutation identity [3], which carries the majority of the findings elsewhere in the work, has no external positive control and does not read these cells. A crystal row does not transfer to it. Quoting the crystal agreement as evidence that the departure measure works moves a result between two instruments that share no part of their construction.

5.5 WHAT A CELL EDGE IS WORTH HERE

Nothing to crystallography. The cells are published to more decimal places than this reaches, by refinement procedures built for the purpose, and a cell recovered from a deposited cell is a round trip and not a new measurement.

What it is worth is the direction of the evidence. An instrument that finds a period nobody told it about, on hundreds of axes whose answers were written down in advance by other people, is displayed to find structure that is there. A memoryless control can only show the instrument does not invent structure that is not. Those are different claims and only one of them can be earned with a memoryless control.

5.6 THE OPEN QUESTION

The detector reads a periodicity. A crystal is the case where periodicity is exact and known. It is the easiest possible instance and it says nothing about the hard one: how far the period has to degrade before the detector stops finding it. Disorder, thermal displacement, a superlattice, an incommensurate modulation and a quasicrystal are all cases where the answer is interesting and none of them is measured here.

That experiment is worth running against published structures where the degree of disorder is itself published, which retains the answer somebody else's number.

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